

Understanding SVM Kernels: How Kernel Choice Shapes Decision Boundaries

Results and Analysis with Visualisations

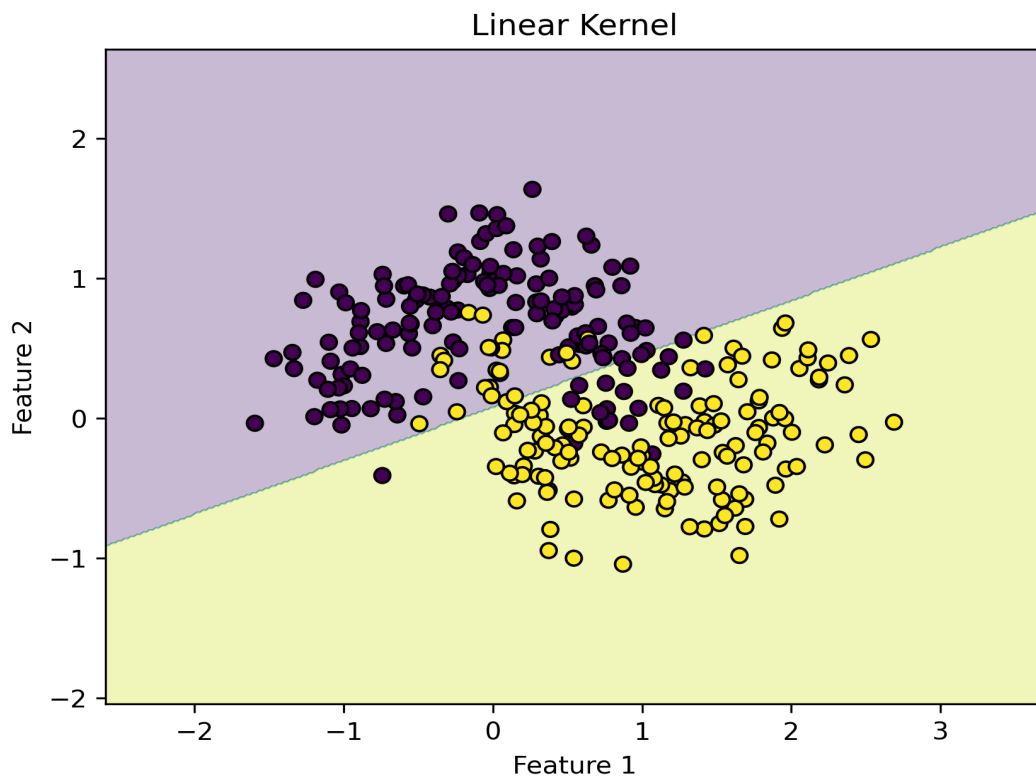


Figure 1: Linear kernel produces a straight decision boundary and underfits the data.

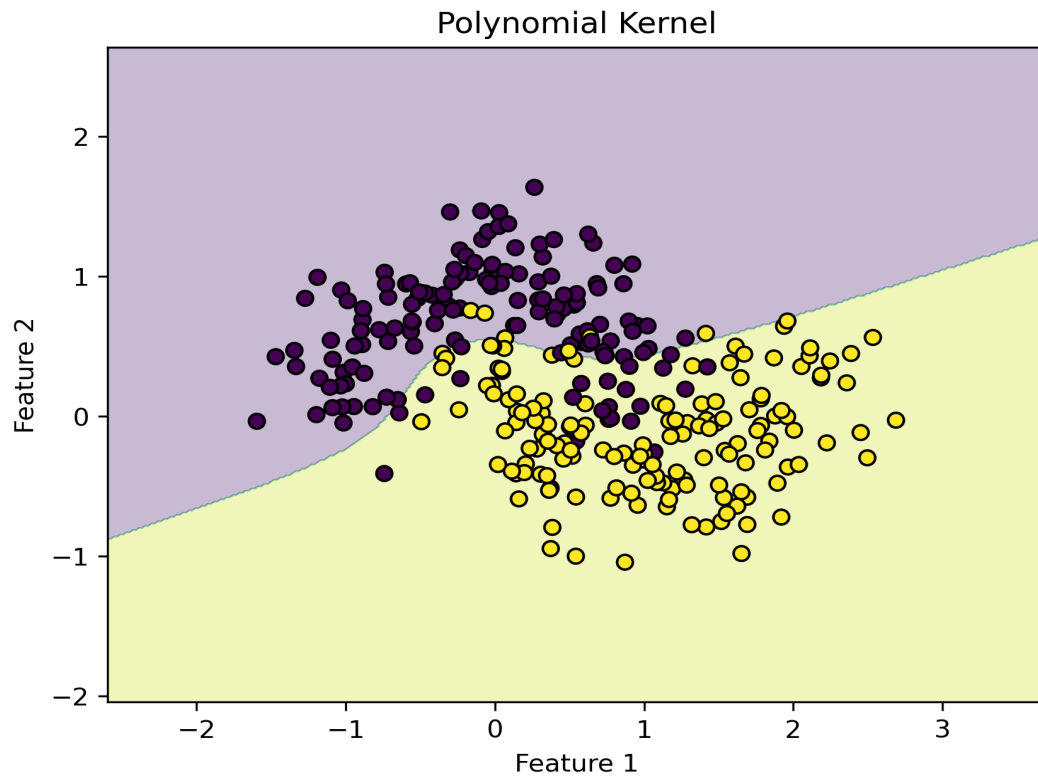


Figure 2: Polynomial kernel introduces curvature and captures more complex patterns.

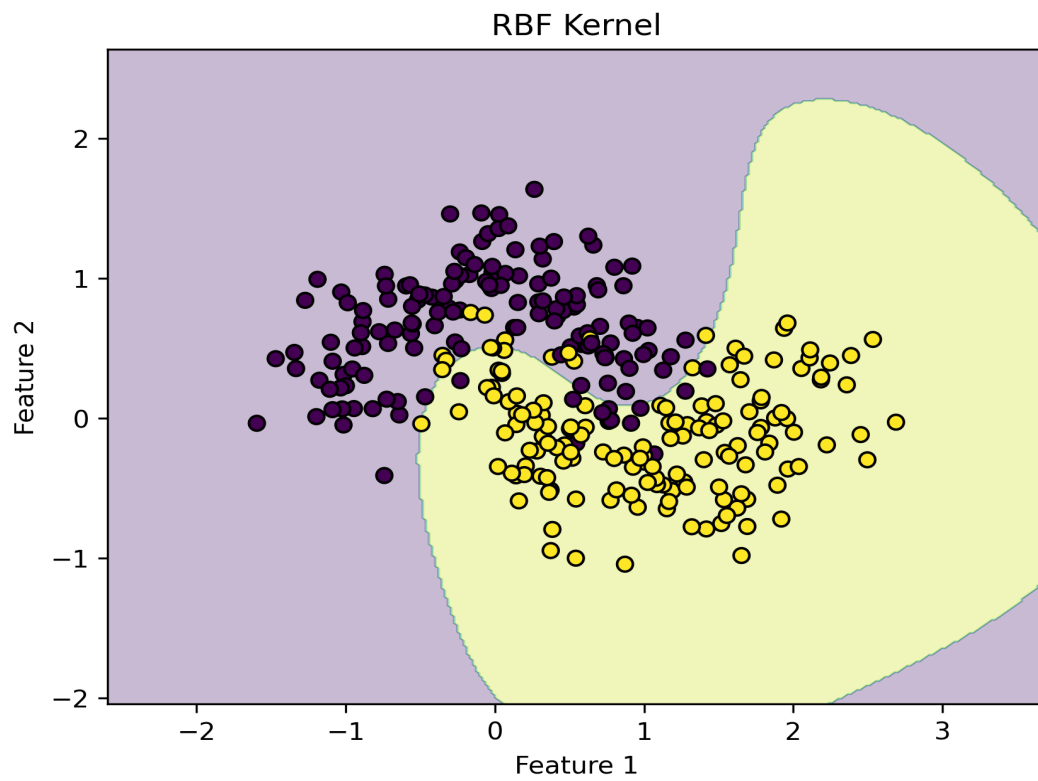


Figure 3: RBF kernel effectively models non-linear patterns and provides the best fit.

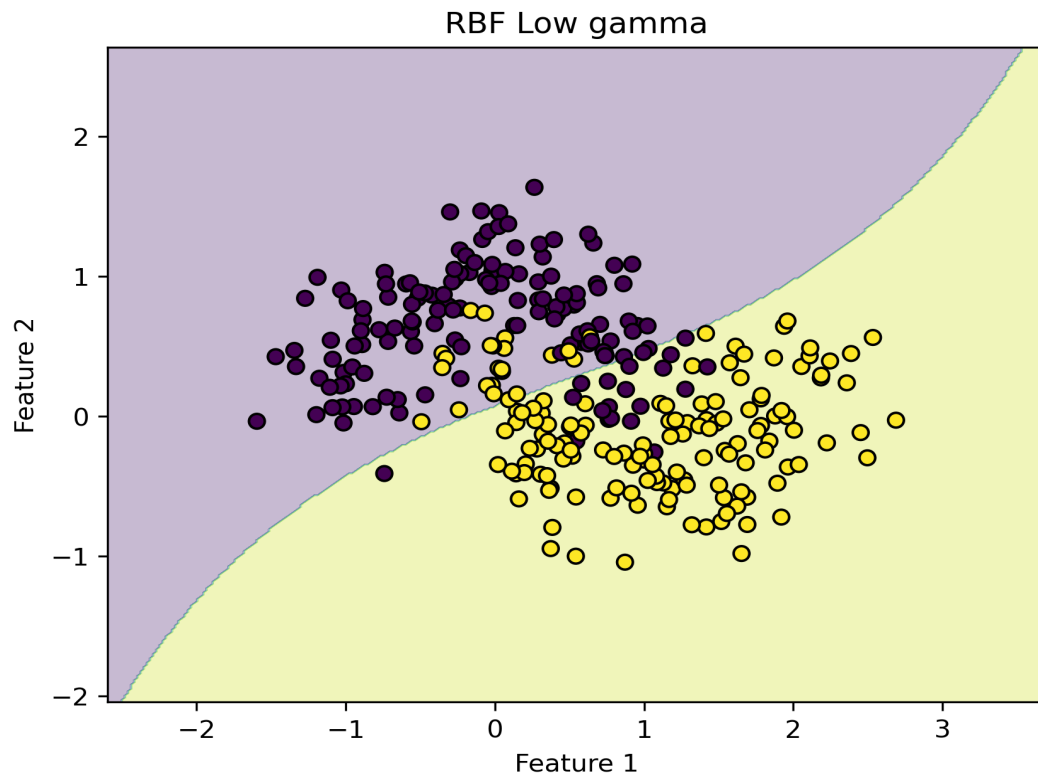


Figure 4: Low gamma results in smoother decision boundaries.

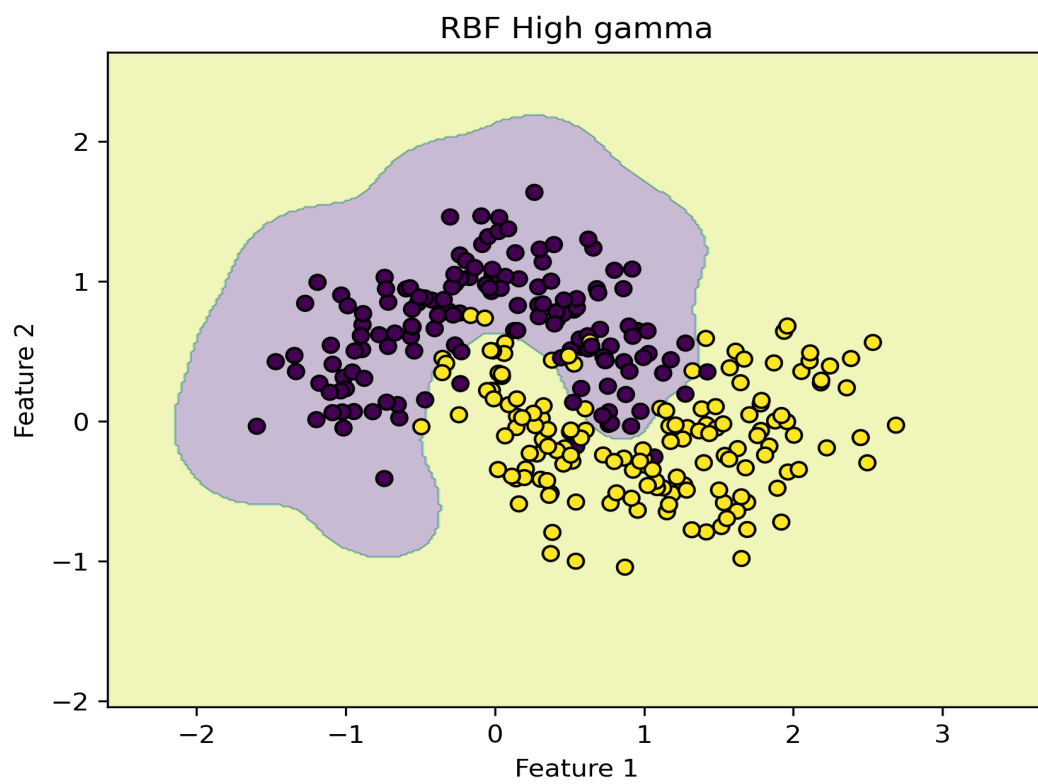


Figure 5: High gamma creates highly complex boundaries and risks overfitting.

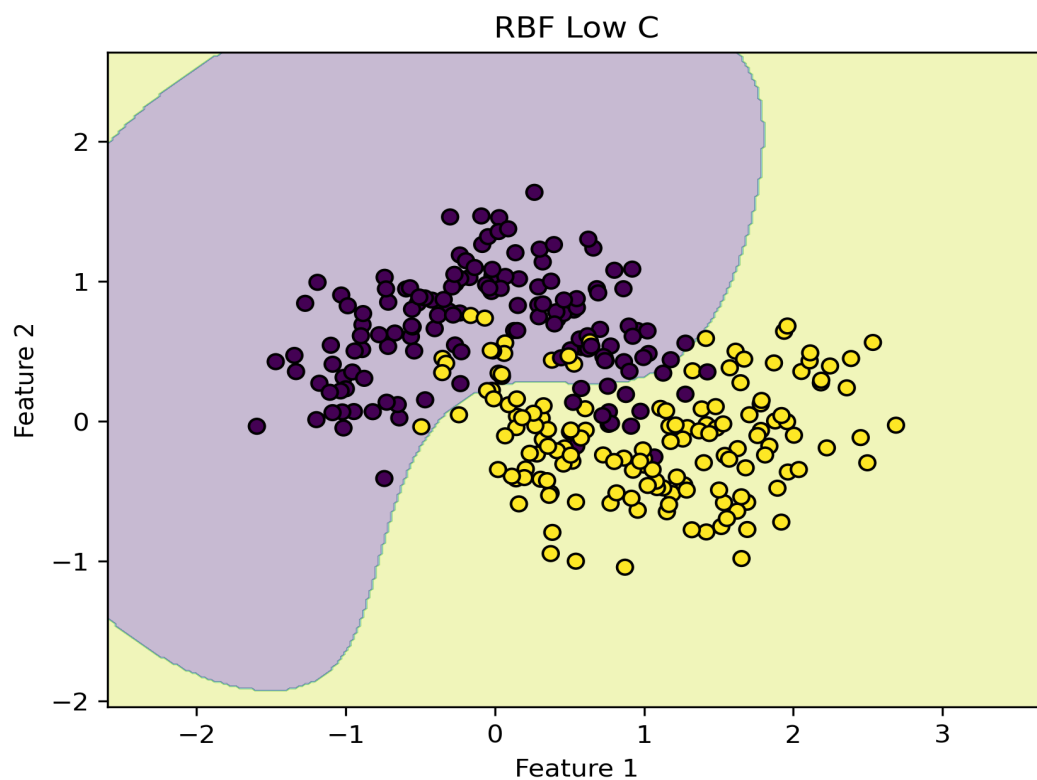


Figure 6: Low C allows wider margins but more misclassification.

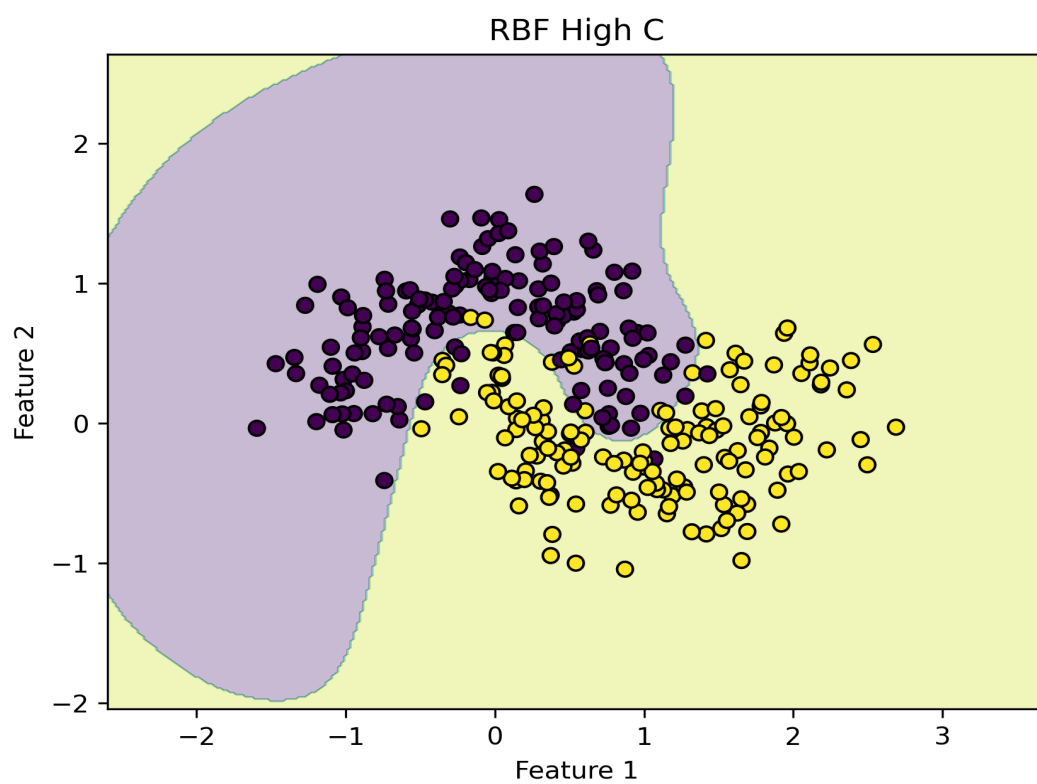


Figure 7: High C reduces error but can lead to overfitting.

References

Cortes, C., Vapnik, V. (1995). Support-vector networks.

Scikit-learn documentation: <https://scikit-learn.org>

Murphy, K. (2012). Machine Learning: A Probabilistic Perspective.